

A Temporal Analysis of Cassini/VIMS Limb Brightness Profiles of Titan from 2004-2017

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ABSTRACT

We analyze the evolution of brightness profiles from *Cassini* Visual and Infrared Mapping Spectrometer (VIMS) spectral mapping cubes to gain insights into the vertical structure of Titan’s atmosphere. Using low-phase angle observations from 2004-2017, we observe changes in the behavior of limb brightening transects with regard to wavelength and time. These transects can independently track the development of different atmospheric layers on a global level. We show that Titan’s limb brightening depends on wavelength, and vary with time. A comparison of brightness profiles from the northern hemisphere and southern hemisphere supports existing analyses of asymmetry behavior. Limb observations of Titan’s upper atmosphere presage existing observations of the lower stratospheric dichotomy, as characterized by Vashist et al. (2023). By using a combination of both observations, we are able to develop a greater understanding of the underlying seasonal dynamics of Titan’s haze, specifically its vertical structure.

Keywords: Titan, Upper atmosphere, Seasonal phenomena

1. INTRODUCTION

Titan’s thick nitrogen atmosphere has proven to be strongly dynamic, responding to seasonal insolation trends in a complex manner. In addition to clouds (Rodriguez et al., 2009; Schaller et al., 2009; Griffith et al., 2009; Brown et al., 2010; Rodriguez et al., 2011a,b; Turtle et al., 2018) and dust storms (Rodriguez et al., 2018) in the lower atmosphere, Titan’s upper atmospheric circulation varies seasonally as evidenced by variations in its haze aerosols (Rannou et al., 2002; Tomasko et al., 2005; Seignovert et al., 2021).

From the Voyager 1 encounter in 1980, Smith et al. (1981) showed a north-south hemispheric dichotomy in the observed haze brightness. Except for distant Voyager 2 imaging (Smith et al., 1982), Titan remained unresolved until the Hubble Space Telescope (HST) based observations were again able to pick up on haze variations (Caldwell et al., 1992; Lorenz et al., 1997, 1999, 2001; Young et al., 2002; Lorenz et al., 2004; Ádámkovics et al., 2006; Karkoschka, 2016, 2022). With adaptive optics, Titan’s hemispheric dichotomy can be measured from ground-based observatories as well (Gibbard et al., 2004; Ádámkovics & de Pater, 2017; Nichols-Fleming et al., 2021).

The arrival of Cassini into the Saturn system in 2004 provided a new opportunity to monitor Titan repeatedly and from close range. The Imaging Science Subsystem (ISS) instrument, a fine-resolution visible-light camera, observed extensive variations of the haze with latitude and time (de Kok et al., 2010; Snell & Banfield, 2024). Cassini’s Visual and Infrared Mapping Spectrometer (VIMS), an imaging spectrometer, mapped haze distributions across a factor of

10 in wavelength from 0.4 – 5.2 μm . VIMS also identified seasonal variations (Seignovert et al., 2021; Vashist et al., 2023) along with diagnostic spatial features (Kutsop et al., 2022).

The different haze appearance, including the hemispheric dichotomy, results from a non-obvious interplay between haze concentration, haze single-scattering albedo, atmospheric absorption, and the vertical structure of the variation of those parameters. For instance, while the north-south dichotomy remains, the brightness ratio between the hemispheres can be different when viewed at different wavelengths — in the visible versus in the near-infrared for instance (Vashist et al., 2023). While the evidence of the vertical variation in haze and atmosphere may be represented in Titan spectra (Rannou et al., 2010), retrievals of atmospheric profiles from spectra alone remain non-unique.

Cassini employed a few different techniques that isolate the haze’s vertical structure. In particular, ISS was able to monitor conditions of the “detached haze” — a thin, discrete high-altitude layer of haze enhancement first seen by Voyager 1 (Smith et al., 1981). Lavvas et al. (2009) and Seignovert et al. (2017) constrained properties of the aerosol particles within a particular detached haze layer. The haze dropped sharply in altitude from ~ 500 km down to ~ 370 km at the time of Titan’s northern spring equinox in 2008 (West et al., 2011). The detached haze later disappeared in 2012 and reappeared later in 2016 West et al. (2018).

The other technique that reveals vertical structure uses occultations. When a star, or the Sun, passes behind the limb of Titan as viewed from Cassini, the spectral lightcurve of the occultation depends on vertical atmospheric variations. VIMS observed a set of 14 relevant occultations (Sotin et al., 2016; Robinson et al., 2014), from which atmospheric absorptions (Bellucci et al., 2009) and haze abun-

83 dance could be ascertained at those particular times and
 84 places (Kim & Courtin, 2013; Maltagliati et al., 2015; Ran-
 85 nou et al., 2022). Observations of stellar occultations in
 86 ultraviolet wavelengths (Koskinen et al., 2011) showed in-
 87 tricate structure unable to be adequately fit by simple haze
 88 parameterizations (Tribbett et al., 2021).

89 We suggest that it may be possible to use limb darkening
 90 across Titan’s spatially resolved disk at wavelengths with
 91 differing degrees of gaseous absorption to probe haze prop-
 92 erties as a function of altitude. In this paper, we report ob-
 93 served limb darkening or limb brightening parameters for
 94 Titan’s haze throughout the Cassini mission, leaving de-
 95 tailed analysis and modeling for future efforts. We discuss
 96 our observations from the VIMS instrument in Section 2,
 97 and the methods that we use to analyze those observations
 98 as well as a simple two-layer model for their interpretation
 99 in Section 3. We then discuss the results of the analysis in
 100 Section 4 before concluding in Section 5.

101 2. OBSERVATIONS

102 As shown in Table 1, we analyze Cassini VIMS-V and
 103 VIMS-IR observations from 24 targeted flybys and 5 non-
 104 targeted flybys, spanning the Cassini probe’s 13 year op-
 105 erational period from 2004-2017. The VIMS instruments
 106 have been well characterized Brown et al. (2004), with the
 107 VIMS-V (Visual) channel of the using a two-dimensional ar-
 108 ray detector covering a spectral range from 0.30 to 1.05 μm
 109 with 96 distinct bands and the VIMS-IR (Infrared) channel
 110 using a one-dimensional detector covering a broader spec-
 111 tral range of 0.3 to 5.1 μm with 256 wavelength bands.

112 2.1. Cube Selection

113 We selected flybys using the following criteria: (1) simul-
 114 taneous observations from both the VIMS-V and VIMS-IR
 115 instruments; (2) a low phase angle ($\leq 30^\circ$ preferred); (3) full-
 116 disk coverage with high limb visibility; (4) sufficient spatial
 117 resolution (200km/pixel); (5) sub-spacecraft longitude be-
 118 tween 20°S and 20°N ; and (6) sufficient time cadence so
 119 as to obtain a comprehensive distribution of measurements
 120 during the *Cassini* mission. It is important to note that
 121 during the latter half of the *Cassini* mission, the spacecraft
 122 was in a polar orbit, which limited the number of flybys
 123 that met the above criteria, leading to the selection of non-
 124 targeted flybys, expansion of criteria, and sparser temporal
 125 coverage.

126 3. METHODS

127 3.1. SRTC++

128 3.2. Image Processing

129 Calibrated VIMS-V data was processed to remove any
 130 variations in vertical pixel arrangements. Since all cubes
 131 were full disks with limb visibility, destriping was able to be
 132 performed using the sky pixels as a baseline from each ver-
 133 tical pixel line, eliminating a majority of the inconsistencies
 134 in the transects **cite the paper that talks about VIMS-**
 135 **V destriping here.** All post-processed data was human
 136 validated through a comparison of the original and pro-
 137 cessed images. There was no post-processing on calibrated
 138 VIMS-IR cubes, nor on any SRTC++ cubes. Known sur-
 139 face bands from **paper to cite on known Titan surface**

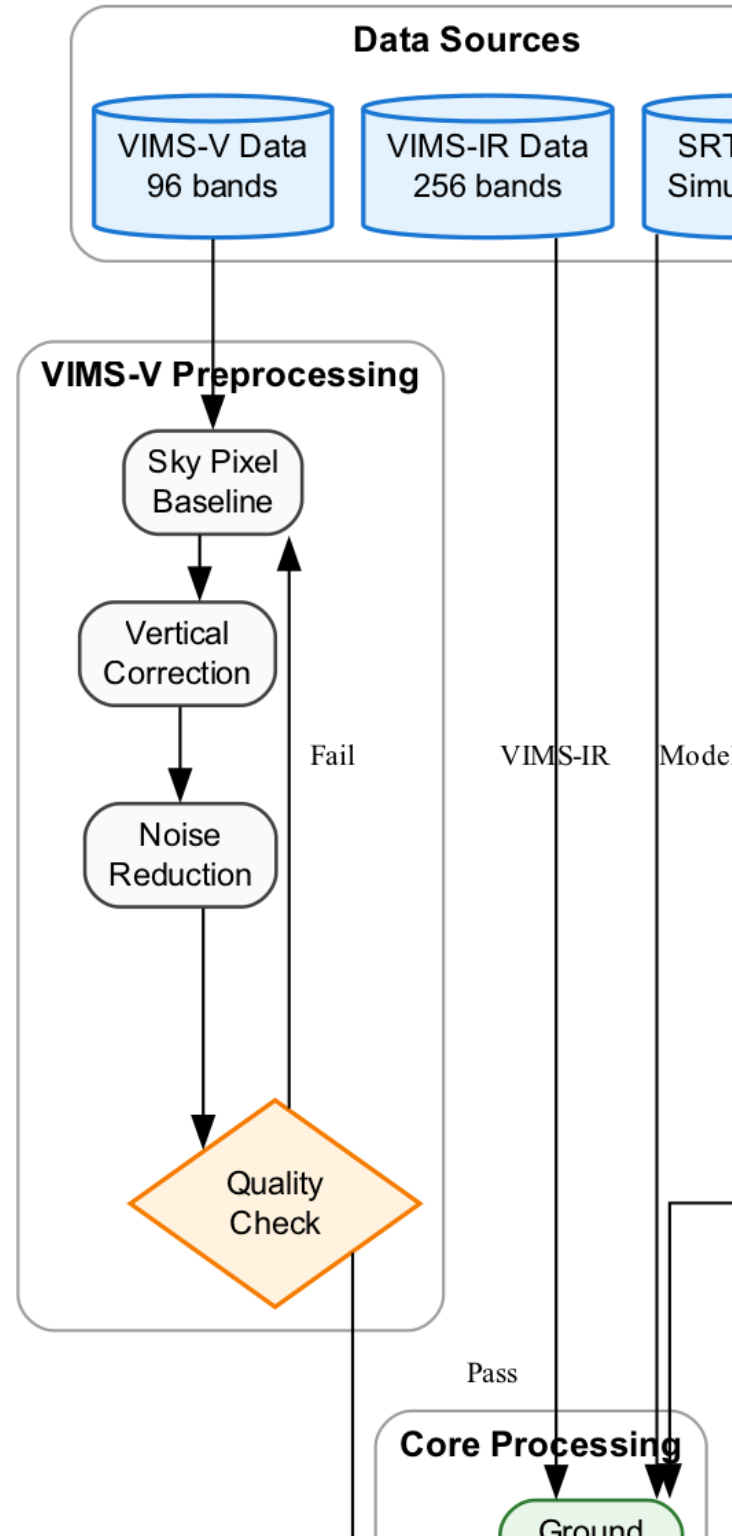


Table 1. North-South Boundary Observations

Year	Month	Cassini flyby	Sub-Spacecraft Latitude (°)	Spatial Resolution (km/pixel)	Phase Angle (°)
2004	10	TA	-15 N	108 km/pixel	13
2005	2	T3	-3 N	147 km/pixel	20
2005	10	T8	n/a	79 km/pixel	23
2005	12	T9	n/a	82 km/pixel	28
2006	2	T11	n/a	100 km/pixel	18
2007	5	T30	15 N	137 km/pixel	28
2007	5	T31	10 N	87 km/pixel	23
2007	6	T32	2 N	109 km/pixel	15
2007	6	T33	n/a	134 km/pixel	12
2007	8	T35	-3 N	127 km/pixel	27
2007	10	051TI ¹	14 N	207 km/pixel	26
2009	7	T58	-28 N	116 km/pixel	28
2009	8	T61	-7 N	142 km/pixel	14
2009	10	T62	-1 N	145 km/pixel	11
2010	4	T67	n/a	88 km/pixel	16
2011	4	T75	n/a	124 km/pixel	16
2011	6	T77	n/a	89 km/pixel	22
2011	12	T79	n/a	124 km/pixel	17
2012	1	T81	-8 N	107 km/pixel	23
2012	5	T83	-8 N	116 km/pixel	23
2012	6	T84	-14 N	119 km/pixel	28
2013	5	191TI ¹	-9 N	267 km/pixel	28
2014	4	T100	50 N	130 km/pixel	35
2015	7	T112	n/a	117 km/pixel	26
2015	11	T114	-1 N	99 km/pixel	26
2016	1	T115	-1 N	120 km/pixel	27
2016	12	255TI ¹	46 N	326 km/pixel	20
2017	5	273TI ¹	38 N	243 km/pixel	16
2017	6	278TI ¹	53 N	179 km/pixel	28

¹ non-targeted flyby of Titan

window wavelengths were not included in the analysis to avoid contamination of the haze signal. The complete data processing workflow is illustrated in Figure 1.

Limb profiles were sampled from flyby observations of Titan using a transect method. The center of the disk for each cube was determined by locating where the line normal to the stellar surface is collinear with the line of sight of the observer within each cube and determining where the emission angle would be 0. To maximize sampling density, images were upsampled 5x. Transects were sampled for various angles relative to North, aiming to sample the northern and southern hemispheres. As shown in Figure 2, transects are selected based on the angle relative to the equator and the direction relative to East/West. We select two transects, one sampling the Northern hemisphere 30° North of the equator and the other 30° South of the equator. The choice of East/West facing transects is determined based on the viewing geometry, sampling data on the hemisphere that does not contain the terminator. Functionally, this is

determined based on the hemispheric location of the point with the lowest angle between the incoming sunlight and the normal line to the planetary surface. Given that the latitude of the atmospheric dichotomy is rarely the same as the solid-body equator, transect data that is within a pixel of the North-South Boundary location, as determined by **cite the paper that shows this (maybe ours??)**, is removed, including any data points between the boundary and the center of the disk. The result is two transects for each band of each VIMS-V and VIMS-IR cube.

3.3. Quadratic Limb Darkening Law Regression

To determine the magnitude of limb darkening and brightening within each transect, we fit the data using the known stellar Quadratic Limb Darkening Law (QLDL) from Kopal (1950); Brown et al. (2001).

$$\frac{I(\mu)}{I(1)} = [1 - u_1(1 - \mu) - u_2(1 - \mu)^2] \quad (1)$$

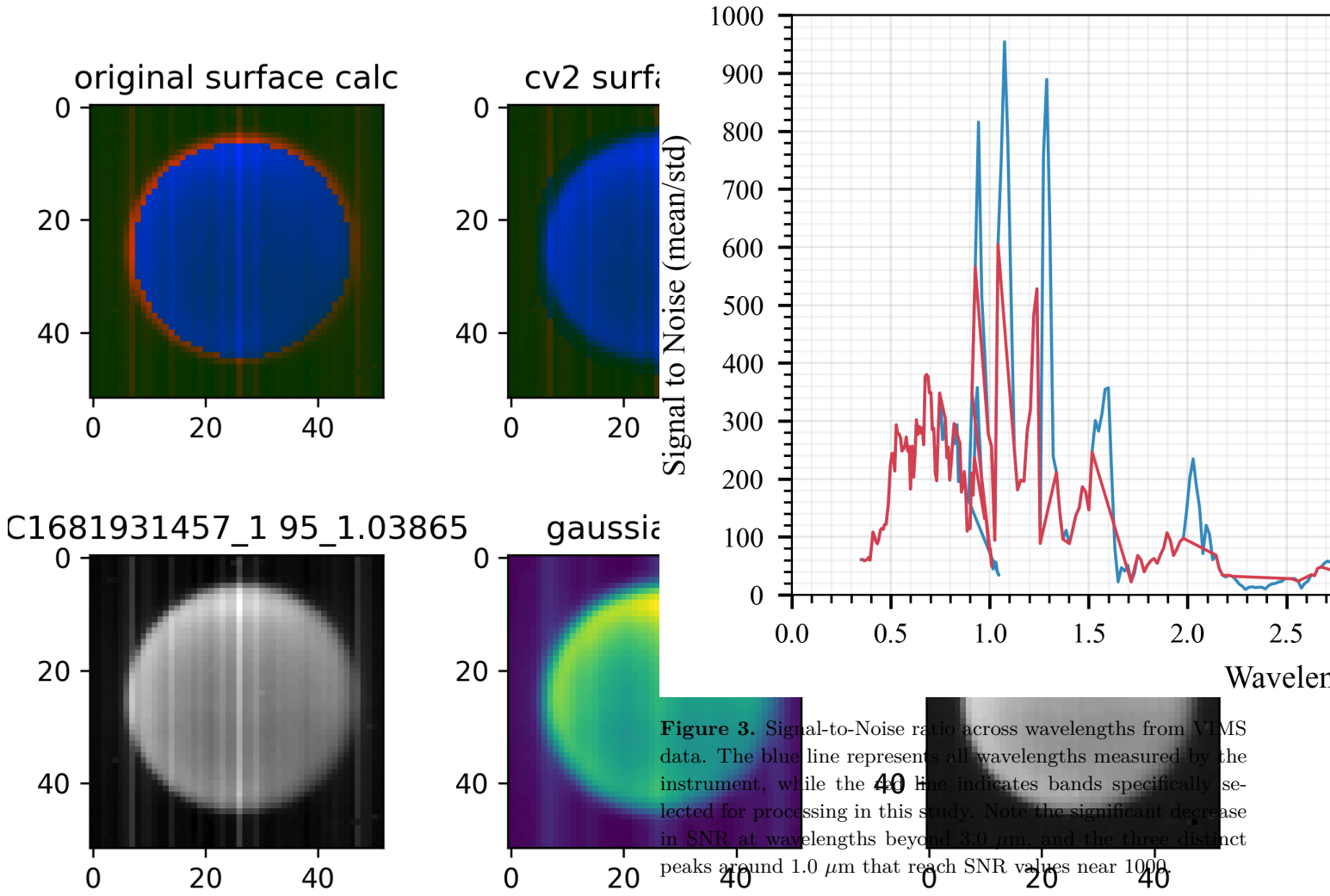


Figure 2. Example of image processing steps for Titan observation C1681931457_1. Top row: (left) original surface calculation showing vertical striping artifacts, (center) improved surface calculation using CV2, (right) plot showing destriping effect with the original data and pixel-by-pixel correction (pchip). Bottom row: (left) raw image with vertical striping, (center) gaussian blur effect, (right) final destriped image showing improved quality for limb analysis.

Figure 3. Signal-to-Noise ratio across wavelengths from VIMS data. The blue line represents all wavelengths measured by the instrument, while the red line indicates bands specifically selected for processing in this study. Note the significant decrease in SNR at wavelengths beyond 3.0 μm , and the three distinct peaks around 1.0 μm that reach SNR values near 1000.

Where μ is the cosine of the angle between a line normal to the stellar surface and the line of sight of the observer leading to a scale from 1 to 0, one representing the center of the disk and 0 being the limb. $I(1)$ represents the intensity at the center of the disk. u_1 and u_2 are limb darkening coefficients. The limb darkening coefficients alone are not well constrained. So to measure the magnitude of limb darkening/brightening, we use $\frac{1}{I(1)} \frac{dI}{d\mu} \Big|_{\mu=0.5} = u_1 + u_2$. Regression used a Levenberg-Marquardt algorithm **maybe cite?** with no parameter bounds applied, though initial values for positive values indicate limb darkening, while negative values indicate limb brightening. Since the sampling density decreases with emission angle, limb pixels gained increased weighting. Resulting fits are compared to the original data in Figure 4. Variations in accuracy are largely attributed to noise in the raw data and lower sampling density at higher emission angles.

Visually, the limb darkening is prevalent at lower, visible, wavelengths, while the limb brightening is more prominent at higher, near-infrared, wavelengths, though the wavelength where darkening and brightening transition seemingly evolves. The differences in limb darkening and brightening are logical when factoring the low albedo of haze at visible wavelengths and the increased haze-ratio at higher altitudes. Limb observations also reinforce existing knowledge of the north-south asymmetry's visual shift, with a

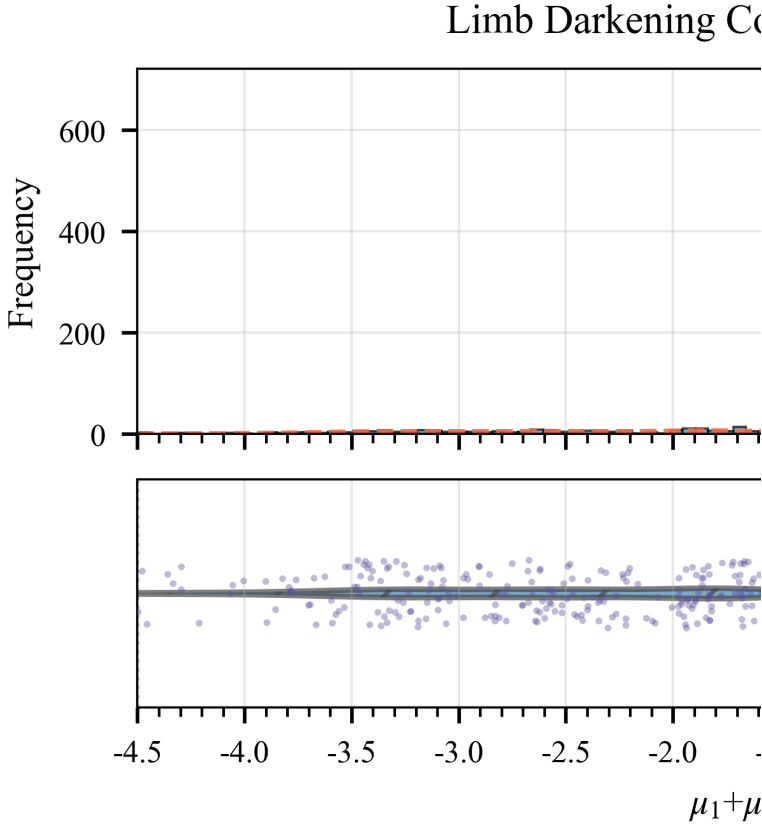


Figure 4. Distribution of limb darkening coefficients ($\mu_1 + \mu_2$) across all analyzed observations. Top: Histogram showing frequency distribution with a peak around 0.0-0.5. Bottom: Violin plot showing the density distribution of all data points, with the majority clustered in the positive range (limb darkening) but with significant extension into negative values (limb brightening). The vertical red dashed line marks the transition between limb darkening (positive values) and limb brightening (negative values). Statistical parameters: Mean = -0.32, Median = 0.01, $\sigma = 1.03$.

201 strong hemispheric dichotomy seen in most bands, as shown
202 in Figure 3.

3.4. Compiled Analysis

4. RESULTS

205 Figure 5 shows the evolution of limb darkening/brightening coefficients ($\mu_1 + \mu_2$) as a function of wavelength for northern and southern hemispheres across three representative time periods of the Cassini mission. Several key features are evident: (1) a consistent transition from limb darkening to limb brightening around 0.75-1.0 μm wavelength, (2) pronounced hemispheric differences that evolve seasonally, and (3) complex wavelength-dependent behavior in the near-infrared region (1.5-3.0 μm) that varies between observations. Early in the mission (2004.82, top panel), both hemispheres show similar patterns but with stronger limb darkening in the south at visible wavelengths. By mid-mission (2007.41, middle panel), greater divergence between hemispheres emerges, particularly in the infrared region where the northern hemisphere shows reduced limb brightening. Late in the mission (2015.51, bottom panel), the trend reverses, with the northern hemisphere showing stronger limb darkening in the visible and the southern

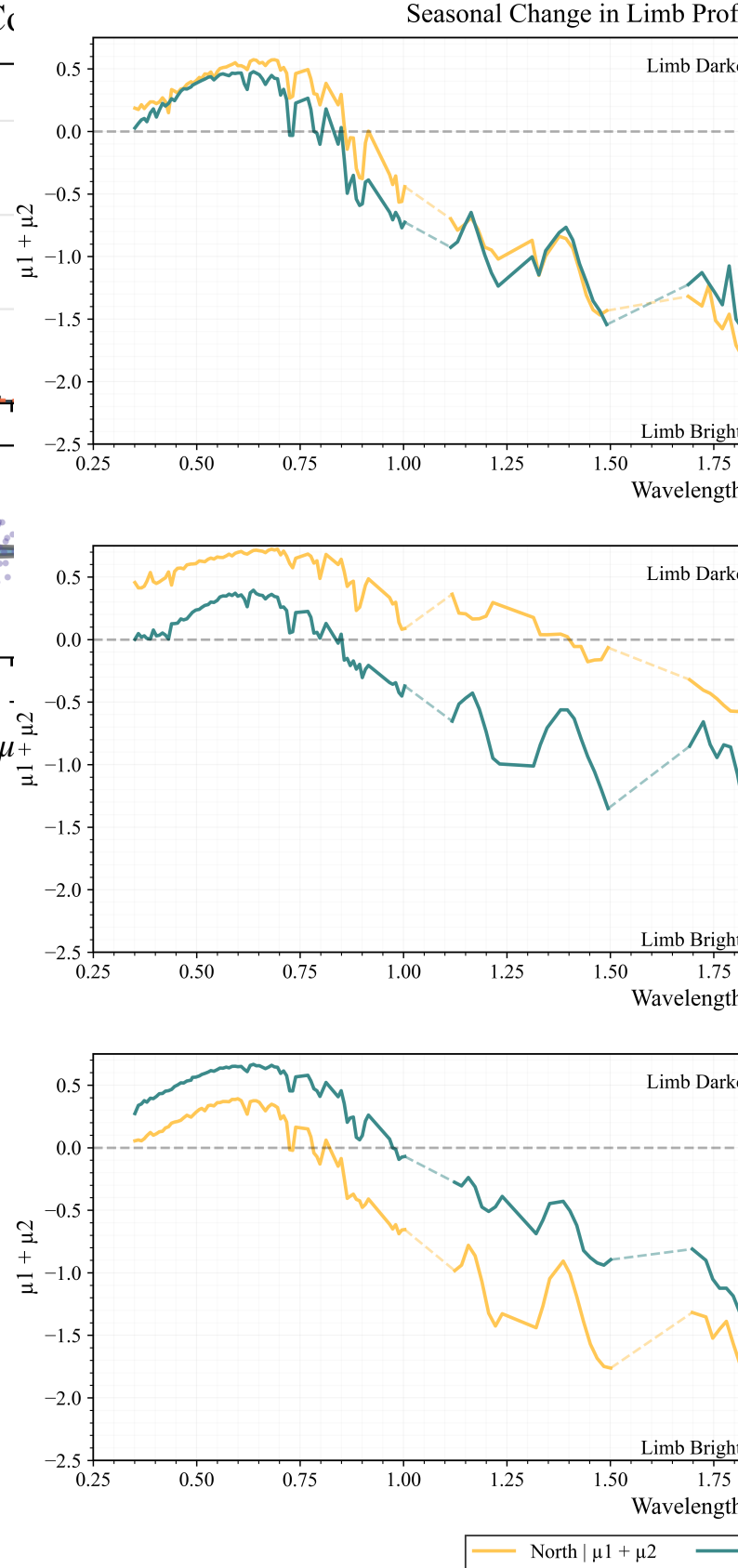


Figure 5. Seasonal changes in limb profiles ($\mu_1 + \mu_2$) vs wavelength (0.25-3.25 μm) for three representative observations spanning the Cassini mission. Top: C1477456872.1 (2004.82), Middle: C1559103132.1 (2007.41), Bottom: C1814906506.1 (2015.51). Each panel shows northern hemisphere (yellow) and southern hemisphere (teal) measurements with dashed trend lines. Positive values indicate limb darkening while negative values indicate limb brightening. Note the wavelength-dependent transition from limb darkening to limb brightening around 0.75-

hemisphere exhibiting more pronounced limb brightening
in the infrared.

5. CONCLUSION

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